



Agaram
TECHNOLOGIES

CASE STUDY

GLOBAL BIOPHARMACEUTICAL ORGANIZATION

A global leader in biopharmaceuticals, specializing in life-saving therapies for rare genetic diseases, needed a solution to enhance clinical data capture from analytical instruments and seamlessly integrate it into their systems such as Plant Historian, MES, etc., for accurate analysis and reporting.

To address these challenges, they adopted Agaram's [InterFacer Middleware](#), enabling seamless integration between instruments, software, and data platforms. By automating the data transfer process, it eliminated the need for manual data entry, reducing the risk of human error.

This transformed laboratory operations by improving data accuracy, efficiency, and scalability, streamlining data collection, and providing real-time access to critical data from instruments like Blood Gas Analyzers, enhancing research and production workflows.



WHAT'S UP WITH BIOTECH LABORATORIES

A global biotech organization specializing in the development of therapies for rare genetic diseases faced significant challenges in collecting and managing data from the analytical instruments used in its research and production processes. Their challenges included,

Legacy Data Collection Process: Data from various analytical instruments were maintained in the legacy system, which increased the risk of errors and inefficiencies, resulting in time delays in reporting and reduced accuracy.

Lack of Integration: Instruments across the labs were not integrated into a unified system, creating silos of data that were difficult to correlate and analyze in real-time.

Data Accuracy and Timeliness: Inaccuracies in data capture and delays in reporting due to manual interventions resulted in inefficient workflows and slower decision-making processes.

Scalability Concerns: With the organization's rapid global expansion and increasing complexity of its research, the existing system lacked the scalability to handle growing volumes of data from new instruments and systems.

Calibration and QC Bottlenecks: Routine calibrations and QC checks performed in BGA systems periodically could not be tracked centrally, limiting traceability.

HOW [INTERFACER MIDDLEWARE](#) INTEGRATES THE LAB

Q: What if you could automatically extract data from analytical lab instruments and integrate it into your systems?

A: InterFacer Middleware is designed to enable seamless integration between laboratory instruments, software systems, and data management platforms, facilitating streamlined, error-free data transfer.

From the start, we have found Agaram Technologies to be very open, informative, and professional in their dealings. These qualities, in addition to their flexible and easy-to-use InterFacer Middleware software, made our decision much easier to choose their product. With Agaram Technologies' expertise, we have been able to integrate our required manufacturing analytical instruments onsite in a reliable manner and their solution also allows us to maintain the integrity of the data being captured.

Automation Projects Lead

Customer Testimonial

GET THE PICTURE

The biotech organization utilized a combination of manual and semi-automated methods for data collection. Data from multiple analytical instruments such as blood gas analyzers, mass spectrometers, and chromatography devices were maintained in standalone legacy systems and databases. There was no direct integration between Plant Historian and MES. This process was not only time-consuming but also prone to human error, leading to inconsistencies in data reporting and analysis.

ROLE OF THE BLOOD GAS ANALYZER (BGA)

A critical component of laboratory setup was the Blood Gas Analyzer (BGA), specifically the RAPIDPoint 500 system, used for point-of-care testing. The system transmits patient sample data, quality control (QC) sample data, and system calibration data to LIMS or to a hospital information system (HIS). This system is essential for measuring parameters like blood gases (pH, pCO₂ i.e. Carbon Dioxide, pO₂ i.e. Oxygen Tension), electrolytes, metabolites, total hemoglobin, and hemoglobin derivatives from whole blood samples (arterial, venous, or capillary).

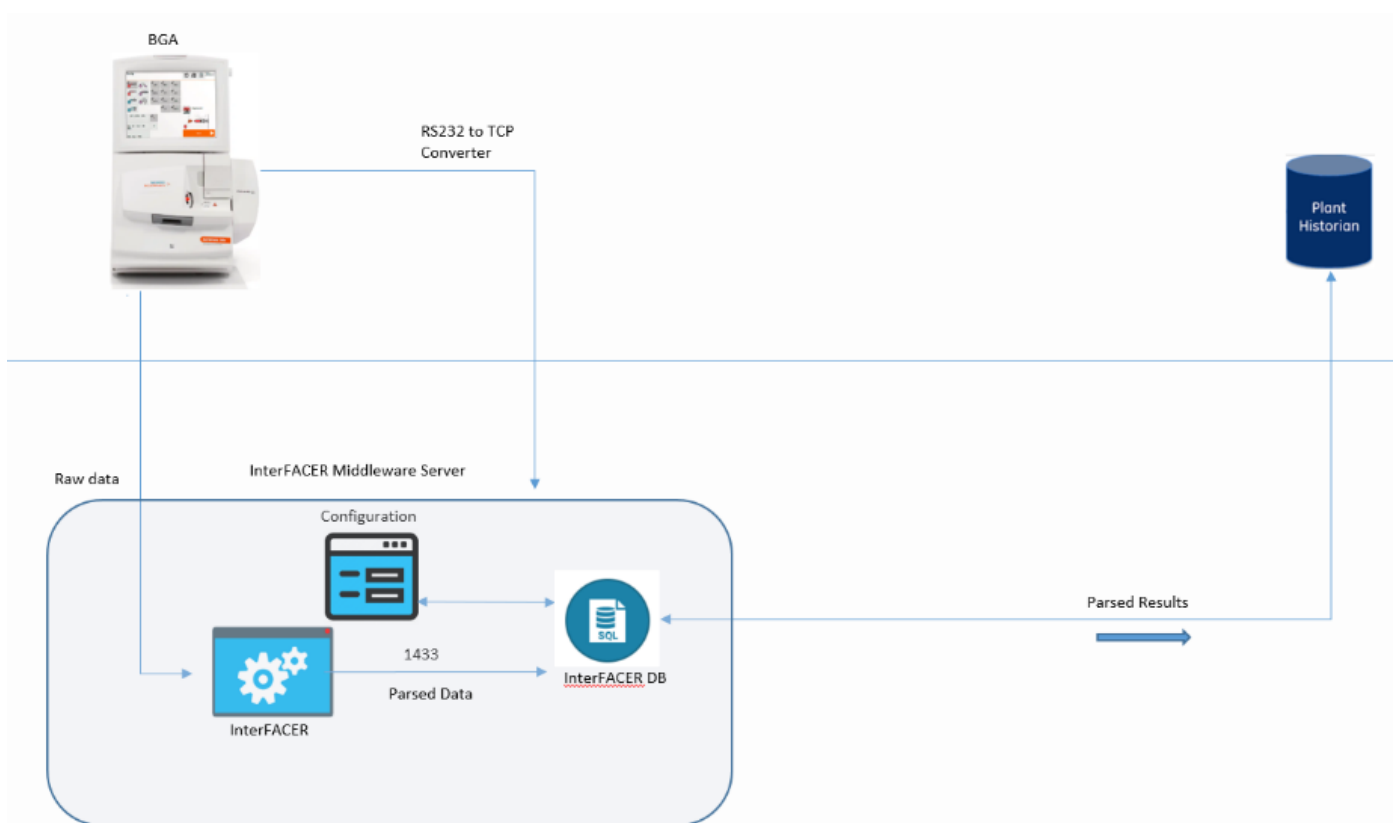
The BGA transmits:

- Patient sample data
- Quality control (QC) sample data
- System calibration data

These are fundamental for real-time diagnostics, process monitoring, and calibration accuracy. However, due to the lack of direct integration with the databases, these critical data points were often delayed or required manual intervention.

AUTOMATED BIOTECH LABORATORIES

The biotech organization needed a robust, real-time integration solution to capture and structure critical data from BGA and analytical instruments, ensure continuous connectivity, automate test result transfers to SQL, and integrate with systems like Plant Historian, MES, ERP, and DeltaV/Siemens PLCs. Thus, they adopted Agaram's **InterFacer Middleware** as it met all their requirements.



InterFacer Middleware acts as a bridge that facilitates communication between various data sources and systems, ensuring that data can flow smoothly without manual intervention. The solution was customized to meet the specific needs of their laboratories, enabling them to capture, process, and store data automatically. It can support OPC (Open Platform Communications) protocol, enabling standardized, real-time data exchange between diverse analytical instruments and control systems in biotech labs. OPC ensured seamless integration of devices like BGA units, spectrometers, and chromatographs into central systems.

REQUIREMENT GATHERING & ANALYSIS

The implementation of Agaram's InterFacer Middleware solution began with a thorough analysis of the organization's existing systems and workflows. Workshops were conducted with key stakeholders from their laboratory teams to gather detailed requirements and identify pain points in the current data capture process. This phase involved:

- Mapping existing systems and understanding the flow of data.
- Identifying the specific instruments and systems that required integration.
- Defining technical and operational requirements for the new system.

SYSTEM VALIDATION & CUSTOMIZATIONS

Based on the gathered requirements, Agaram's team designed a customized solution to integrate the organization's instruments with their existing IT infrastructure. The system architecture was carefully planned to ensure that it could scale with future needs. Custom connectors were built for various instruments to ensure smooth data transfer.

TESTING & VALIDATION

Once the system was developed, extensive testing was conducted to ensure the solution met the organization's requirements. This included:

- **Functional Testing:** Ensuring that the InterFacer Middleware solution captured data accurately from each instrument.
- **Performance Testing:** Verifying that the system could handle high volumes of data without delay.
- **User Acceptance Testing (UAT):** Involving the laboratory team to test the system in real-world scenarios and ensure it met their expectations.

VALIDATION PROCESS

Installation Qualification (IQ)

Agaram Technologies ensured the successful installation of the InterFacer Middleware through a thorough Installation Qualification (IQ) process. The key components of the system, including the required OS (Windows Server 2012 or higher), .NET 4.5 framework and other required components, were verified for proper installation. The services necessary for the system's operation were also confirmed to be installed and running correctly on the designated servers. All required files were found to be present in the specified directories, ensuring the system was set up properly for the subsequent Operational Qualification phase.

Operational Qualification (OQ)

Once the installation was completed, the InterFacer system underwent an extensive Operational Qualification (OQ) phase to confirm that it was fit for its intended use. The tests verified the functionality of the system in real-world conditions, ensuring seamless integration with the organization's instruments, data capture from devices like the Blood Gas Analyzers (BGA), and accurate data transmission to the central SQL database. The system passed all functional tests, including validation of data accuracy, real-time integration, and performance under high data

volumes. The OQ protocol also ensured that the system's features met regulatory compliance standards, providing the organization with a fully validated solution for automating their data capture process.

Discrepancy Management and Resolution

During the IQ and OQ phases, Agaram Project Team conducted a meticulous review of each test to identify any discrepancies. Any discrepancies from the expected results were promptly documented and investigated. For example, in cases where system behaviors did not match the expected results, such as issues with file paths or service startup errors, a detailed investigation was carried out. Corrective actions were taken, including reconfiguration of settings or reinstallation of components, to ensure that the system met the specified requirements. Discrepancies were resolved in collaboration with the organization's team, ensuring that the system was validated without any remaining issues.

TRAINING, GO-LIVE & SUPPORT

After successful testing, the system was deployed in the biotech organization's labs, and the InterFacer Middleware solution went live. Agaram Technologies provided training to their teams to ensure they were comfortable using the new system.

Agaram provided continuous support following the go-live phase to ensure that the InterFacer solution maintained its operational excellence. This included regular system checks and validation of data flows to ensure the continued accuracy and integrity of the data captured from the laboratory instruments. In the months following implementation, the system was monitored for performance, and any potential issues were addressed promptly. This post-implementation support ensured that the laboratory's operations continued to run smoothly, with the system adapting to future expansion without any disruptions.

KEY BENEFITS

- **Increased Efficiency:** Automated data capture eliminated manual BGA entry, reducing time spent on repetitive tasks and allowing lab staff to focus on higher-value activities, boosting operational efficiency.
- **Enhanced Data Accuracy:** Automation minimized human error, ensuring that all data used for analysis and reporting remained accurate, consistent, and reliable.
- **Real-Time Access to Data:** Teams gained real-time visibility into data from all instruments, supporting faster decision-making and timely completion of testing and analysis.
- **Improved Data Integrity:** BGA test results, calibration events, and QC logs were captured accurately and in real time, maintaining the integrity and completeness of records.
- **Scalability for Growth:** As operations expanded and new instruments were introduced, the InterFacer Middleware scaled effortlessly with minimal configuration and no workflow disruption.
- **Improved Regulatory Compliance:** Traceable, verifiable data capture ensured compliance with regulatory standards, improving audit readiness and adherence to industry requirements.
- **Critical Role of OPC:** OPC enabled seamless communication between instruments and central systems, ensuring real-time data exchange, accuracy, and regulatory compliance.

This successful implementation of InterFacer Middleware not only modernized laboratory operations but also laid a strong digital foundation for future growth, positioning the global biopharmaceutical organization to scale efficiently, maintain regulatory compliance, and drive continued innovation in the biotechnology sector.

Curious to see the product in action? Request a demo and get a detailed overview [here](#).

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